

Cold Flow Test Platform

Cold flow testing is how you find out whether a propulsion system's plumbing behaves before there is anything combustible in it. That only works if you can see what happened, which makes the instrumentation as much of a design problem as the structure holding it.

ROLE	SENSOR CHANNELS	DATA PIPELINE	TEST CAMPAIGN
Flight Vehicles Engineer; test vehicles lead	12 (pressure, thermal, vibration)	Python, all channels on a common clock	Vertical integration and cold flow

INSTRUMENTATION AS THE DESIGN PROBLEM

I designed a vertical integration and cold flow test platform that captures pressure, thermal, and vibration data from twelve individual sensors, and wrote the Python pipeline that synchronizes and processes those multi-channel streams into dashboards the team can actually read.

Raw multi-channel DAQ output is close to useless in a debrief. The value is in getting every channel onto a common clock and putting the anomaly in front of the person who needs to see it.

WHERE IT STANDS

The platform has been through a vertical integration and cold flow campaign, with the vehicle standing on the rail, propellant loading infrastructure alongside, and the data acquisition cabinet running the twelve channels. I am continuing as test vehicles lead, running a team of four engineers to instrument and fly the subscale test vehicle.

TEAM

Leading four engineers on test vehicles: instrumenting the subscale canard vehicle and carrying the cold flow data pipeline forward into flight.

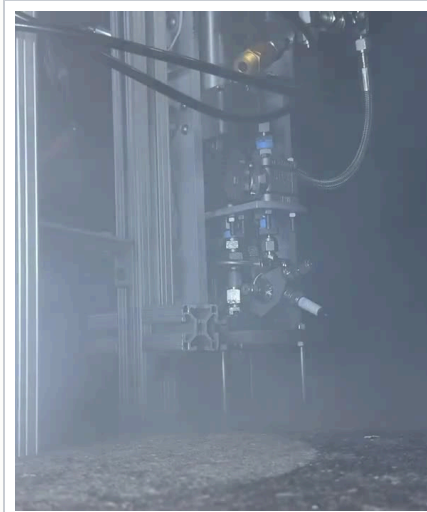


FIG 2 Cold flow, filmed at the base of the stage. The vapor cloud is the working fluid venting through the feed system.



FIG 3 Launch.



FIG 1 The cold flow setup for a night test. The vehicle stands vertically on the rail at the left, propellant loading infrastructure next to it, and the data acquisition cabinet on the far right.

12	4
SENSOR CHANNELS	ENGINEERS LED

TOOLS Ansys Mechanical · Python · DAQ instrumentation

RESEARCH · GÜDER RESEARCH GROUP, IMPERIAL COLLEGE LONDON · JUN-AUG 2026

Passive Thermal and Fluid Control for an Infrastructure-Free Diagnostic Cassette

Holding a DNA amplification reaction at temperature for twenty minutes normally takes a heater, a controller, and a power source, which is exactly the equipment a point-of-care test exists to eliminate. This one runs on a salt.

PRINCIPAL INVESTIGATOR Prof. Firat Güder, Bioengineering	REQUIREMENT 37–42 °C held for 10 min, zero external power	OPERATOR INPUT One water activation step	MY ROLE Owned the project end to end: design, fabrication, characterization, analysis
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A TWO-STAGE PASSIVE SYSTEM

An annular heating chamber wraps concentrically around the central sample well, maximizing heated wall area against the reaction volume and keeping the thermal path short and radially symmetric. The sample well sits directly upstream of a lateral flow strip, separated by a sacrificial membrane that sets when amplified product is released.

- Exothermic charge.** Water activation triggers an MgCl_2 hydration exotherm: high energy density and fast release, but an uncontrolled spike on its own.
- PCM buffer.** A phase-change ring absorbs the exotherm peak as latent heat, then returns it slowly on crystallization, turning a spike into a usable plateau.
- Sacrificial valve.** A cast PVA membrane swells and ruptures on a fixed timescale, releasing product to the strip only after amplification completes.

WHAT I DID

Designed the cassette geometry in Fusion 360 and iterated the printed body through several revisions of the heating chamber. Built the characterization rig with a thermocouple sealed into the sample well so every run reported reaction-volume temperature, not chamber-wall temperature. Swept PCM mass against MgCl_2 mass across 49 combinations and wrote the Python/Matplotlib pipeline that reduced raw traces to minutes-in-window. Separately cast PVA membranes across concentration, cast volume, and plasticizer, screening rupture time and failure mode with tracer dye on real sample pads.

RESULT

The best configuration (1800 mg PCM / 500 mg MgCl_2) held the reaction volume inside 37–42 °C for about 11 minutes with no electrical input. A limited end-to-end trial amplified the *mecA* resistance gene from MRSA inside the cassette, confirmed by gel electrophoresis.

Open items: the optimum sits on a narrow ridge, so charge-mass tolerance matters; the valve shows build-to-build timing scatter; and 11 minutes is short of the 20 the protocol wants.

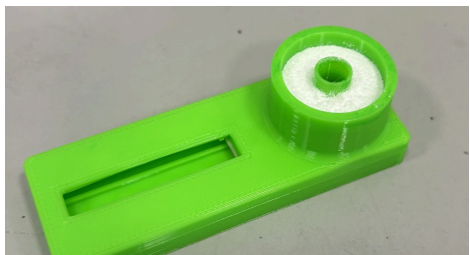


FIG 1 Printed cassette body. The annular heating chamber surrounds the central sample well; the lateral flow assay sits below.

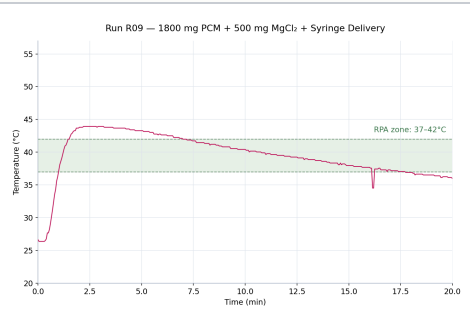


FIG 2 Best configuration. The charge drives the volume to ~44 °C within two minutes; the PCM then returns heat slowly, carrying the volume through the 37–42 °C window for ~11 min.

PCM mg \ MgCl_2 mg	250	300	350	400	450	500	550
1000	0.80	1.10	0.90	1.50	1.80	1.90	1.90
1200	0.40	0.90	2.50	3.40	4.60	4.00	2.60
1400	0.70	1.30	2.90	4.60	6.90	6.10	5.00
1600	0.80	1.60	2.40	4.60	7.97	9.53	8.60
1800	1.10	1.50	2.10	3.90	7.93	11.13	10.47
2000	1.10	1.50	1.30	2.40	5.70	8.90	9.90
2200	0.60	1.20	0.60	1.70	3.20	5.00	6.30

FIG 3 Minutes in window across the 7 × 7 sweep. Non-monotonic in both parameters; the operating point had to be found empirically.

TOOLS Fusion 360 · FDM printing · membrane casting · thermocouple logging · Python · Matplotlib · gel electrophoresis

RESEARCH · INTERDISCIPLINARY MICROSYSTEMS GROUP, UNIVERSITY OF FLORIDA · AUG 2025-PRESENT

Lateral Filter Array Microfluidics for Circulating Tumor Cell Isolation

03

There are a handful of tumor cells in a tube of blood containing billions of everything else. Finding them is a separation problem, and the geometry of the channel is what does the separating.

PRINCIPAL INVESTIGATOR Prof. Z. Hugh Fan, Mechanical & Aerospace Engineering	MODELLING COMSOL Multiphysics	FABRICATION & READOUT Mask design, soft lithography, fluorescence microscopy, ImageJ	VALIDATION Clinical samples, UF Health Shands Hospital
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FIG 1 Lateral filter array device, dye-filled, penny for scale. The serpentine main channel carries the bulk of the sample; the comb structures branching off it are the filter arrays where larger cells are trapped off the flow path.

RESEARCH FOCUS

Circulating tumor cells are shed into the bloodstream by solid tumors, and capturing them from a routine blood draw offers a route to cancer diagnosis and monitoring that does not require a biopsy. The difficulty is abundance: the target cells are vanishingly rare against a background of red and white blood cells, so any capture method has to be both highly selective and gentle enough that the cells survive to be analysed.

The lateral filter array device addresses this with geometry. A serpentine main channel carries the bulk of the sample while an array of side filters, sized between the diameters of a blood cell and a tumor cell, traps the larger cells off the main flow path. Because the filters sit lateral to the flow rather than across it, the device resists the clogging that limits conventional membrane filtration.

MY RESPONSIBILITIES

I built a COMSOL model of the microfilter array relating channel size and volumetric flow rate to capture efficiency, so that a device could be designed toward a target rather than found by iteration. I then fabricated and tested more than 200 microfluidic devices to validate that model, working through mask design, soft lithography, and fluorescence microscopy for readout, and cross-checking predicted against measured capture using clinical samples supplied by UF Health Shands Hospital.

WHY THE READOUT MATTERS

Every captured object has to be classified: DAPI-positive, panCK/CSV-positive and CD45-negative is a tumor cell; CD45-positive is a white blood cell that made it through the filter. Deciding between the two on every object is what makes capture efficiency a number worth measuring.

200+
DEVICES
FABRICATED &
TESTED

COMSOL
MODEL VALIDATED
ON CLINICAL
SAMPLES

A

Merged

CD45

DAPI

panCK+
CSV-FITC

BF

B

Merged

CD45

DAPI

panCK+
CSV-FITC

BF

FIG 2 Immunofluorescence readout. A: DAPI+, panCK/CSV-FITC+, CD45-, the signature of a circulating tumor cell. B: CD45+, a leukocyte that passed the filter.

TOOLS

COMSOL Multiphysics · mask design · soft lithography · fluorescence microscopy · ImageJ